Department of Information Technology
Academic Year 2024 - 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT
Course Code & Title: CS3352 & Foundations of Data Science
Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit IV **/ Python Libraries- Numpy and Pandas**
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO8
- **Activity Chosen:** Think Pair Share
- **Justification:**
  In unit IV, Numpy and Pandas package is an important topic needed for every data scientist because the data analysis done by using this package only. This activity helps the students to know the implementation of package functions.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
  - The topic Python Libraries Numpy and Pandas taught in the previous days.
  - The time allotted for the topic is 15 Minutes.
  - The students were asked to form a team with two pairs. The total teams are 31, the questions are allotted to each team as shown in figure 1,2.
  - Asked the student to discuss with their pairs and find the answer for their corresponding questions. The time allotted is 10 minutes.
  - The students were asked to present their solution as shown in figure 3.
  - First Priyadharshini and Jeyalakshmi presented and they discuss the answer for reshaping an array questions. Naveen kumar and Vignesh presented the answer for concatenation and splitting an array.
  - Next Vijayakumar and Manish Vardhan team presented initializing multi dimensional array initialization as shown in the Figure 4, in the same way all the teams were presented.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	2	2	1	2	2	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO9**	**PO10**	**PO12**	**PSO1**	**PSO2**	**PSO3**
	3	3	2	2	2	1	2	2	3	2
Justification for correlation	Apply basic Knowledge and fundamentals of mathematics for operating on data	The students will be able to identify the need for analysis of complex engineering problems	Able to design the process flow for developing data science applications.	Able to analyze the data and represent the data with valid conclusion	The students able to function as a team in analyzing and describing data.	Interpret the packages for data science process	The students able to recognize the need for packages for data science process.	The students able to exhibit the knowledge in python packages	The students can Exhibit knowledge of data science in the field of IoT, Cloud Computing to develop IT solutions	The students able to exhibit knowledge in python packages to develop a quality product.

- **Images / Screenshot of the practice:**

Figure 1: Think Pair Share

Questions:

1. Initializing a multidimensional array using a list of lists
2. Create a 3x5 floating-point array filled with 1s
3. Create a 3x5 array filled with 3.14
4. Create an array filled with a linear sequence
   # Starting at 0, ending at 20, stepping by 2
5. Create an array of five values evenly spaced between 0 and 1
6. Create a 3x3 array of random integers in the interval [0, 10]
7. Accessing subarrays in one dimension and in multiple dimensions
8. x2 = np.random.randint(10, size=(3, 4)) # Two-dimensional array
   print(x2)
   [[12 5 2 4]
   [7 6 8 8]
   [1 6 7 7]]
   Display
   # two rows, three columns
   # all rows, every other column
9. x = np.array([1, 2, 3])
   y = np.array([3, 2, 1])   Concatenate and split two arrays
10. Find absolute value of an array
    x = np.array([-2, -1, 0, 1, 2])
11. compute some trigonometric functions, Exponent, Logarithms
    theta = np.linspace(0, np.pi, 3)
12. Use Reduce on the add, reduce on the multiply in an array
    x = [1, 2, 3, 99, 99, 3, 2, 1]
13. From Dictionary create series
    Create a following dictionary

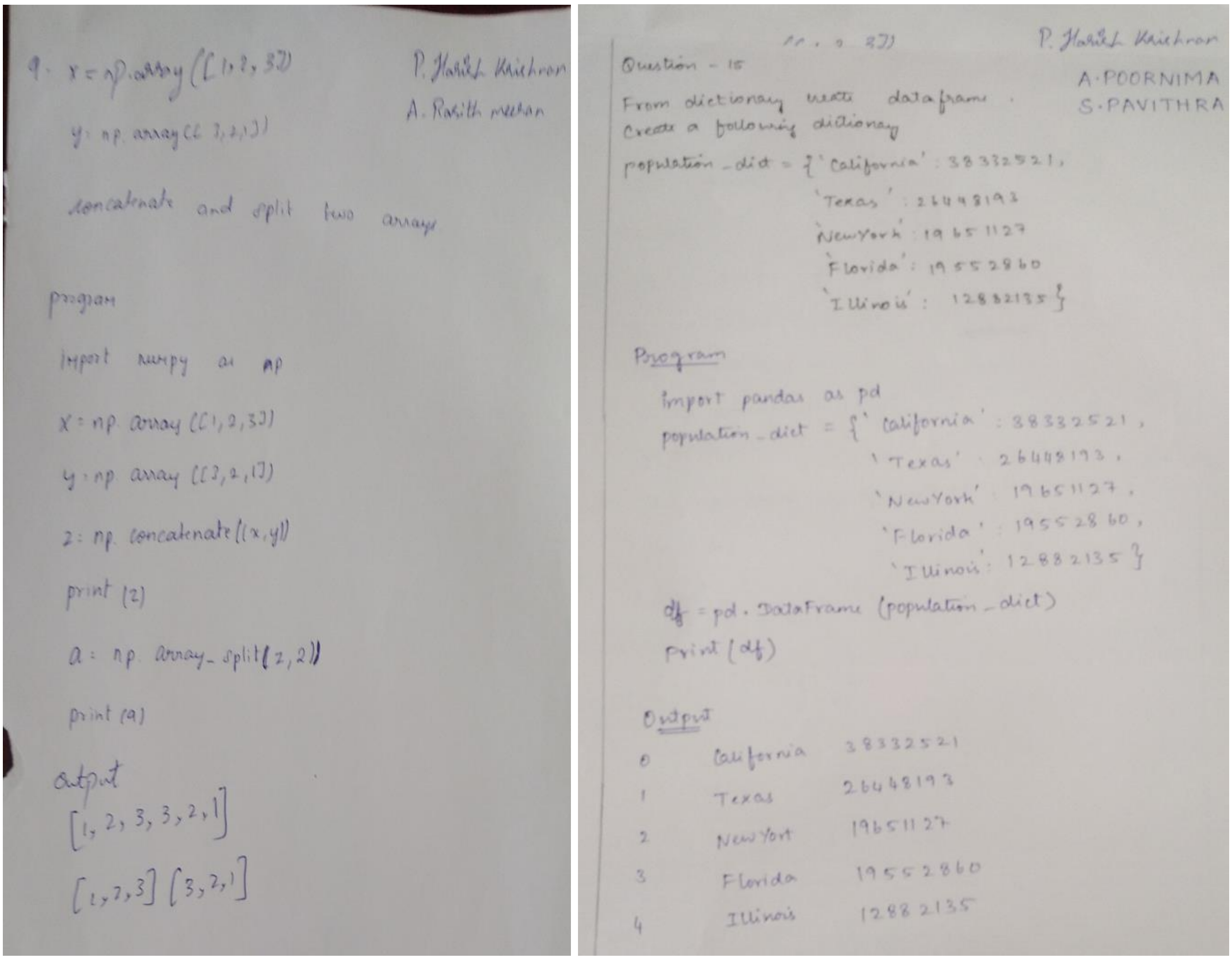
population_dict = {'California': 38332521,'Texas': 26448193,'New York': 19651127,
'Florida': 19552860,
'Illinois': 12882135}

14. area_dict = {'California': 423967, 'Texas': 695662, 'New York': 141297, 'Florida': 170312, 'Illinois': 149995}
15. From Dictionary create dataframe
    Create a following dictionary
    population_dict = {'California': 38332521,
    'Texas': 26448193,
    'New York': 19651127,
    'Florida': 19552860,
    'Illinois': 12882135}
16. From Dictionary create dataframe
    area_dict = {'California': 423967, 'Texas': 695662, 'New York': 141297, 'Florida': 170312, 'Illinois': 149995}
17. Transpose the Dataframe
    Create a following dictionary
    population_dict = {'California': 38332521,'Texas': 26448193,'New York': 19651127,
    'Florida': 19552860,
    'Illinois': 12882135}
18. Use columns and indices in a dataframe
    A = pd.DataFrame(rng.randint(0, 20, (2, 2)))
19. Add A and B Dataframe using Ufunc
    A = pd.DataFrame(rng.randint(0, 20, (2, 2)),columns=list('AB'))
    B = pd.DataFrame(rng.randint(0, 10, (3, 3)),columns=list('BAC'))
20. Create a two-dimensional array with a size 3x2 of data with random values and give the column names as rand1,rand2.
21. Create a Dataframe and perform the following
    data = [{'a': i, 'b': 2 * i}
    for i **in** range(3)]
    pd.DataFrame(data)
22. Create a series and perform the operations using loc and iloc
    data = pd.Series(['a', 'b', 'c'], index=[1, 3, 5])
    data
23. Print columns in the following dataframe
    A = pd.DataFrame(rng.randint(0, 20, (2, 2)),
    columns=list('AB'))
24. Perform reindex in the dataframe
    population_dict = {'California': 38332521,'Texas': 26448193,'New York': 19651127,
    'Florida': 19552860,'Illinois': 12882135}
25. Modify Dataframe object
    population_dict = {'California': 38332521,'Texas': 26448193,'New York': 19651127,
    'Florida': 19552860,'Illinois': 12882135}
26. Use comparison operator in numpy array
    rng = np.random.RandomState(0)
    x = rng.randint(10, size=(3, 4))
    x
27. Apply masking operation for the following array
    rng = np.random.RandomState(0)
    x = rng.randint(10, size=(3, 4))

x

28. Create a series of data and perform the following display values, index.
    data = pd.Series([0.25, 0.5, 0.75, 1.0])
    data
29. Create an array and to find the minimum value and maximum value of any given array
30. Apply horizontal split of an array
    x = [1, 2, 3, 99, 99, 3, 2, 1]
31. Apply vertical split of an array
    x = [1, 2, 3, 99, 99, 3, 2, 1]

Figure 2: Questions



P. Harish Krishnan
A. Rohith mohan

9. x = np.array([1,2,3])
y = np.array([3,2,1])

concatenate and split two arrays

program

```
import numpy as np
x = np.array([1,2,3])
y = np.array([3,2,1])
z = np.concatenate((x,y))
print(z)
a = np.array_split(z,2)
print(a)
```

output

[1, 2, 3, 3, 2, 1]

[1,2,3] [3,2,1]

11.9.21

P. Harish Krishnan
A. POORNIMA
S. PAVITHRA

Question - 15

From dictionary create dataframe.
Create a following dictionary

population_dict = {'California': 38332521,
'Texas': 26448193
'NewYork': 19651127
'Florida': 19552860
'Illinois': 12882135}

Program

```
import pandas as pd
population_dict = {'California': 38332521,
                   'Texas': 26448193,
                   'NewYork': 19651127,
                   'Florida': 19552860,
                   'Illinois': 12882135}
df = pd.DataFrame(population_dict)
Print(df)
```

Output

0	California	38332521
1	Texas	26448193
2	New York	19651127
3	Florida	19552860
4	Illinois	12882135

Figure 3 Solution given by Harishkrishnan and Poornima Team

Figure 4 Presentation by Naveen Kumar and Jeyalakshmi Team

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
    - Students said that they were able to understand the concepts of Python packages
    - The students said that the activity help to apply the packages in different ways.
    - The students were shared their ideas for using the python libraries
    - It makes the students to improve their thinking skills.
- ***Benefit of the practice:***
    - Students can able to recall the different methods and their applications in data science process.
    - The students can share their ideas with their pairs.
    - It helps the slow learners to understand the concepts.
- ***Challenges faced in implementation:***
    - The Activity planned for 15 minutes but it takes 20 minutes to complete the activity.
    - Some students hesitate to present their solution.

References:

- https://www.kent.edu/ctl/think-pair-share
- https://www.readingrockets.org/classroom/classroom-strategies/think-pair-share

Signature of Faculty Member **HOD**